

Microchip DAC Breakout Module (12 Bit)

CHIP: MCP4725A0T-E**SIZE:** 15.5mm x 13mm

Description:

This DAC Breakout Module breaks out GND, VDD, I2C (SDA & SCL) & output voltage and features the MCP4725 series typically operating on 3.3v or 5v. This module features 1uF & 100nF capacitors on VDD, and an LED indicating good power. This board takes out the tedious, unpredictable troubles that come from DAC's, and will let you streamline and debug your inventions & innovations.

PINOUT:

PIN	NAME	DOES THIS
1	GND	Ground
2	SCL	Serial Clock
3	SDA	Serial Data
4	VOUT	Output Voltage
5	VDD	+3.3v (Typical)

Applications:

- Precision voltage generator
- Audio & Waveform Generation
- Prototyping Analog Circuits

CHIPS ON BOARD:

COMPONENT	PART #	JOB	DATASHEET
DAC	MCP4725A0T	Digital to Analog Conversion	[link]

Example Code (ESP32):

```

#include <Wire.h>

#define SDA_PIN 21
#define SCL_PIN 22
#define DAC_ADDR 0x60 // MCP4725A0T, A0 pin tied low

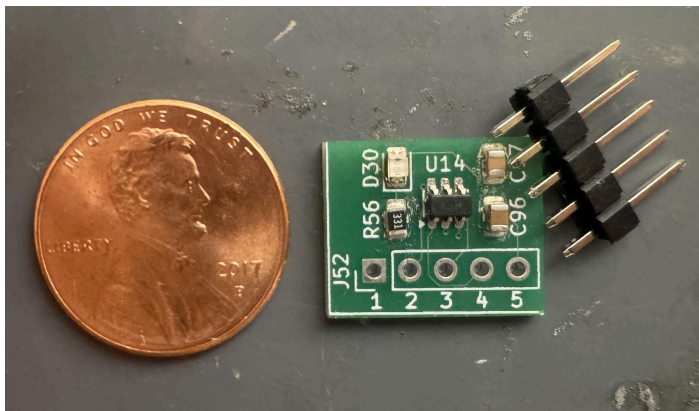
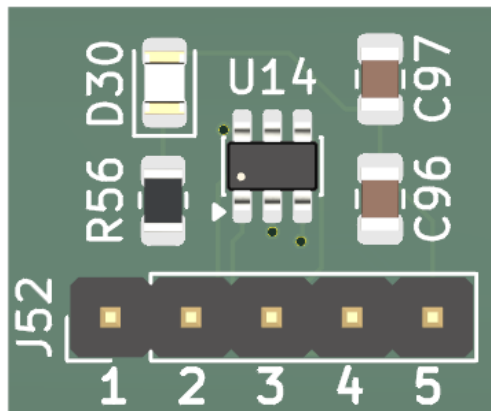
void writeDAC(uint16_t value) {
    value &= 0x0FFF;

    Wire.beginTransmission(DAC_ADDR);
    Wire.write((value >> 8) & 0x0F);
    Wire.write(value & 0xFF);
    Wire.endTransmission();
}

void setup() {
    Serial.begin(115200);
    Wire.begin(SDA_PIN, SCL_PIN);
    Wire.setClock(400000);
    Serial.println("MCP4725 ready (raw I2C). Starting sweep...");
}

void loop() {
    for (uint16_t val = 0; val <= 4095; val += 8) {
        writeDAC(val);
        delayMicroseconds(500);
    }
    for (int16_t val = 4095; val >= 0; val -= 8) {
        writeDAC(val);
        delayMicroseconds(500);
    }
}

```

PHOTOS:

Notes & Debugging:

- This board features a green LED indicating power good. The LED's brightness is high for easy recognizability & debugging. If you wish, you could dim the LED by using a larger resistor, or remove it by unsoldering the LED.
- The MCP4725 features 12 bit resolution, uses I2C communication and operates on a 2.7V to 5.5V range.
- Be mindful of the 1uF and 100nF capacitor on board, to ensure you don't overload your power source (Most commonly a problem in USB's).
- If using the dedicated pin headers shipped with your DAC, proper soldering is needed to ensure full connection on all 5 pins.